

Maximum daily wages of an officer = ₹ 354

Number of Maximum days to attend the duty = $\frac{4956}{354} = 14$

Number of days present on duty = $\frac{3894}{354} = 11$

Number of absent days = $14 - 11 = 3$ days

378. Let, the two natural numbers be a and b

According to given information

$$\therefore a^2 - b^2 = 19$$

$$\Rightarrow (a + b)(a - b) = 19 \times 1$$

$$\Rightarrow a + b = 19 \dots (i)$$

$$a - b = 1 \dots (ii)$$

On adding,

$$2a = 20$$

$$\Rightarrow a = 10$$

From equation (i),

$$\therefore 10 + b = 19$$

$$\Rightarrow b = 19 - 10 = 9$$

$$\therefore a^2 + b^2 = (10)^2 + (9)^2$$

$$= 100 + 81 = 181$$

379. Let the two numbers be a and b

$$\therefore a + b = 14 \dots (i)$$

$$a - b = 10 \dots (ii)$$

By adding equation (i) and (ii) we get

$$2a = 24$$

$$\therefore a = 12 \text{ and } b = 2$$

$$\therefore \text{product of these two numbers} = 12 \times 2 = 24$$

380. Let A is to be added then $2x^2 + 3x - 5 + A = x^2 + x + 1$

$$\Rightarrow A = x^2 - x + 1 - (2x^2 + 3x - 5) \Rightarrow A = x^2 - x + 1 - 2x^2 - 3x + 5$$

$$= -x^2 - 4x + 6$$